PAEDIATRIC ORTHO · NEET-SS / FET / INI-SS

Perthes and SCFE

Perthes disease (idiopathic AVN of the capital epiphysis)

- Age **4-8**, boys 4-5:1, bilateral in about 10-15% (asynchronous), delayed bone age. Symmetrical bilateral changes suggest **multiple epiphyseal dysplasia** or hypothyroidism instead.
- **Waldenstrom stages:** initial (dense epiphysis, **crescent sign**), fragmentation, reossification, healed.

Classification	What it tells you
Herring lateral pillar (fragmentation stage)	A: pillar height normal. B: over 50% preserved. B/C border: about 50%, or a narrow or poorly ossified pillar. C: under 50%. Best predictor of outcome, together with age
Catterall I-IV	Extent of head involvement; "head at risk" signs: **Gage sign**, lateral calcification, lateral subluxation, horizontal physis, metaphyseal cysts
Salter-Thompson	Extent of the subchondral fracture: A under 50%, B over 50%
Stulberg (at maturity)	I-II spherical and congruent (good); III-IV aspherical congruence; **V aspherical incongruence** (worst)

- **Prognosis: age at onset** is the most important factor (under 6 years does well), then lateral pillar class. Girls fare worse.
- **Treatment = containment.** Under 6 years: symptomatic care and range of motion. **Over 8 years with lateral pillar B or B/C:** surgical containment (proximal femoral varus osteotomy

or Salter innominate osteotomy) gives better heads (Herring's multicentre study).
- **Hinge abduction:** valgus-extension osteotomy or shelf acetabuloplasty (salvage).

Slipped capital femoral epiphysis (SCFE)

- Obese adolescents: boys 13-15, girls 11-13. Bilateral in 20-40%.
- Suspect an **endocrine cause** (hypothyroidism, growth hormone therapy, renal osteodystrophy) if under 10, over 16, or thin.
- The metaphysis moves anterior and externally rotated; the epiphysis stays posterior and medial in the acetabulum.
- **Signs:** obligatory external rotation on flexion (**Drehmann sign**); knee or thigh pain (obturator nerve referral). Never miss it in a child with knee pain.
- **Loder stability** (the key prognostic split): **stable** = can walk, AVN risk near 0; **unstable** = cannot walk even with crutches, AVN risk up to about 50%.
- **Imaging: Klein line** does not cross the epiphysis (Trethowan sign); Steel metaphyseal blanch sign. **Southwick slip angle** on the lateral view: mild under 30 degrees, moderate 30-50, severe over 50. The frog-leg lateral is the most sensitive view, but avoid it in an unstable slip.
- **Treatment: in situ fixation with one cannulated screw**, centred in the epiphysis, perpendicular to the physis, with no joint penetration. Do not reduce a stable slip.
- Prophylactic fixation of the other hip: endocrine cause, age under 10, open triradiate cartilage, poor follow-up.
- Severe or unstable slips: **modified Dunn** realignment through a surgical dislocation, in experienced hands.

- **Complications:** AVN (unstable slip, forceful reduction), **chondrolysis** (screw penetration), cam-type FAI.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.